

Примери решених

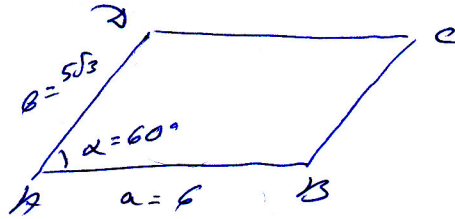
13. ABCD - II-к

$$a = 6$$

$$b = 5\sqrt{3}$$

$$\alpha = 60^\circ$$

$$S = ?$$



$$S = a \cdot b \cdot \sin \alpha$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$S = 6 \cdot 5\sqrt{3} \cdot \frac{\sqrt{3}}{2} =$$

$$= 3 \cdot 5 \cdot 3 = 15 \cdot 3 = 45$$

$$\text{зачува } \sqrt{3} \cdot \sqrt{3} = 3 !!!$$

23. ABCD - II-к

$$a = 9$$

$$b = 12\sqrt{2}$$

$$\alpha = 135^\circ$$

$$S = ?$$

$$\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$S = a \cdot b \cdot \sin \alpha = 9 \cdot 12\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 9 \cdot 12 = 108$$

$$\text{зачува } \sqrt{2} \cdot \sqrt{2} = 2 !!! \text{ и } \frac{2}{2} = 1$$

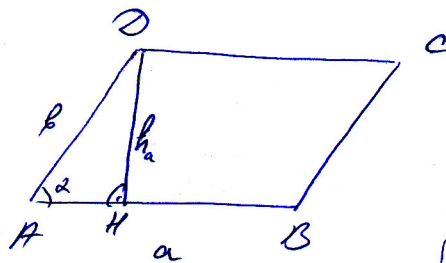
33. ABCD - II-к

$$a = 7$$

$$b = 3\sqrt{2}$$

$$\alpha = 45^\circ$$

$$h_a = ?$$



$$S = a \cdot b \cdot \sin \alpha$$

$$S = a \cdot h_a \Rightarrow h_a = \frac{S}{a}$$

$$\sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$S = 7 \cdot 3\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 7 \cdot 3 = 21 \Rightarrow h_a = \frac{S}{a} = \frac{21}{7} = 3$$

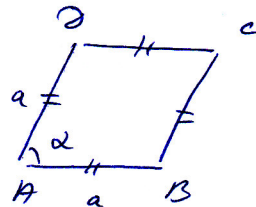
По овим подацима се израчунава $h_a = \frac{S}{a}$

43. ABCD - ромб - това е четуространик с 4 еднаких стране

$$a = 8$$

$$\alpha = 30^\circ$$

$$S = ?$$



$$S = a \cdot a \cdot \sin \alpha$$

$$\sin 30^\circ = \frac{1}{2}$$

$$S = 8 \cdot 8 \cdot \frac{1}{2} = 8 \cdot 4 = 32$$

53. ABCD - 11-к $\boxed{S = a \cdot b \cdot \sin \alpha} \Rightarrow b = \frac{S}{a \cdot \sin \alpha}$

$a = 6$ $\sin 30^\circ = \frac{1}{2}$

$\alpha = 30^\circ$

$S = 21$

$b = ?$

$$b = \frac{S}{a \cdot \sin \alpha} = \frac{21}{6 \cdot \frac{1}{2}} = \frac{21}{3} = 7$$

63. ABCD - 11-к $\sin 45^\circ = \frac{\sqrt{2}}{2}$ $\boxed{S = a \cdot h_a} \Rightarrow h_a = \frac{S}{a}$

$a = 8\sqrt{2}$ $\boxed{S = b \cdot h_b} \Rightarrow h_b = \frac{S}{b}$

$\alpha = 45^\circ$ $\boxed{S = a \cdot b \cdot \sin \alpha} \Rightarrow b = \frac{S}{a \cdot \sin \alpha} = \frac{40}{8\sqrt{2} \cdot \frac{\sqrt{2}}{2}} = \frac{40}{8} = 5$

$S = 40$

$b = ?$ $h_a = ?$ $h_b = ?$

$$h_a = \frac{S}{a} = \frac{40}{8\sqrt{2}} = \frac{5}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{5\sqrt{2}}{2} = 2,5\sqrt{2}$$

$$h_b = \frac{S}{b} = \frac{40}{5} = 8$$

73. ABCD - четырехугольник (т.е. 11-к, ромб, трапеция...)

$d_1 = 4$

$d_2 = 5$

$\varphi = 60^\circ$

$S = ?$

$\boxed{S = \frac{1}{2} d_1 d_2 \sin \varphi}$ $\sin 60^\circ = \frac{\sqrt{3}}{2}$

$$S = \frac{1}{2} \cdot 4 \cdot 5 \cdot \frac{\sqrt{3}}{2} = 5\sqrt{3}$$

83. Δ ABC $\boxed{S = \frac{1}{2} a \cdot b \cdot \sin \gamma}$ $\sin 30^\circ = \frac{1}{2}$

$a = 8$

$b = 11$

$\gamma = 30^\circ$

$S = ?$

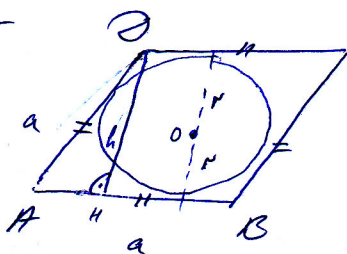
$$S = \frac{1}{2} \cdot 8 \cdot 11 \cdot \frac{1}{2} = 2 \cdot 11 = 22$$

9. ABCD - ромб

$$a = 6$$

$$\alpha = 45^\circ$$

$$r = ?$$



$$S = a \cdot a \cdot \sin \alpha$$

$$\sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$S = a \cdot h \Rightarrow h = \frac{S}{a}$$

$$h = 2r \Rightarrow r = \frac{h}{2}$$

$$S = a \cdot a \cdot \sin 45^\circ = 6 \cdot 6 \cdot \frac{\sqrt{2}}{2} = 6 \cdot 3\sqrt{2} = 18\sqrt{2}$$

$$h = \frac{S}{a} = \frac{18\sqrt{2}}{6} = 3\sqrt{2}$$

$$r = \frac{h}{2} = \frac{3\sqrt{2}}{2} = 1,5\sqrt{2}$$

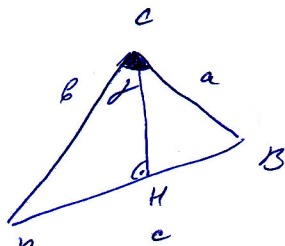
10. $\triangle ABC$

$$a = 3$$

$$b = 2$$

$$\gamma = 60^\circ$$

$$h_c = ?$$



$$S = \frac{1}{2} a \cdot b \cdot \sin \gamma$$

$$\cos 60^\circ = \frac{1}{2}$$

$$S = \frac{1}{2} c \cdot h_c \Rightarrow h_c = \frac{2S}{c}$$

$$\cos \gamma : c^2 = a^2 + b^2 - 2ab \cos \gamma$$

$$S = \frac{1}{2} \cdot 3 \cdot 2 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$c^2 = a^2 + b^2 - 2ab \cos \gamma = 3^2 + 2^2 - 2 \cdot 3 \cdot 2 \cdot \frac{1}{2} = 9 + 4 - 6 = 7$$

$$c^2 = 7 \Rightarrow c = \sqrt{7}$$

$$h_c = \frac{2S}{c} = \frac{3\sqrt{3}}{\sqrt{7}} = \frac{3\sqrt{3}}{\sqrt{7}} \cdot \frac{\sqrt{7}}{\sqrt{7}} = \frac{3\sqrt{21}}{7}$$

113. в ABC

$$p = \frac{a+b+c}{2}$$

$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

$$a = 18$$

$$b = 20$$

$$c = 34$$

$$S = p \cdot r \Rightarrow r = \frac{S}{p}$$

$$S = \frac{abc}{4R} \Rightarrow R = \frac{abc}{4S}$$

$$S = \frac{1}{2} c \cdot h_c \Rightarrow h_c = \frac{2S}{c}$$

$$l_c = \frac{2}{a+b} \sqrt{abp(p-c)}$$

$$m_c = \frac{1}{2} \sqrt{2(a^2+b^2)-c^2}$$

$$S = ? \quad r = ? \quad R = ? \quad h_c = ? \quad l_c = ? \quad m_c = ?$$

$$p = \frac{a+b+c}{2} = \frac{18+20+34}{2} = \frac{72}{2} = 36$$

$$S = \sqrt{36(36-18)(36-20)(36-34)} = 6\sqrt{18 \cdot 16 \cdot 2} = 6 \cdot 4\sqrt{18 \cdot 2} =$$

$$= 24\sqrt{9 \cdot 2 \cdot 2} = 24 \cdot 3 \cdot 2 = 72 \cdot 2 = 144$$

$$r = \frac{S}{p} = \frac{144}{36} = \frac{24 \cdot 3 \cdot 2}{6 \cdot 6} = \frac{24}{6} = 4$$

$$R = \frac{abc}{4S} = \frac{18 \cdot 20 \cdot 34}{4 \cdot 144} = \frac{12240}{576} = \frac{17 \cdot 720}{8 \cdot 72} = \frac{17}{8} = 2,125$$

$$h_c = \frac{2S}{c} = \frac{2 \cdot 144}{34} = \frac{144}{17} = 8 \frac{8}{17}$$

$$\frac{144 \cdot 17 = 8}{136}$$

$$l_c = \frac{2}{a+b} \sqrt{abp(p-c)} = \frac{2}{18+20} \sqrt{18 \cdot 20(36-34)} = \frac{2}{38} \sqrt{18 \cdot 20 \cdot 2} =$$

$$= \frac{1}{19} \sqrt{2 \cdot 9 \cdot 2 \cdot 10 \cdot 2} = \frac{3}{19} \sqrt{4 \cdot 10 \cdot 2} = \frac{3 \cdot 2}{19} \sqrt{10 \cdot 2} = \frac{6}{19} \sqrt{5 \cdot 2 \cdot 2} = \frac{6 \cdot 2\sqrt{5}}{19} = \frac{12\sqrt{5}}{19}$$

$$m_c = \frac{1}{2} \sqrt{2(a^2+b^2)-c^2} = \frac{1}{2} \sqrt{2(18^2+20^2)-34^2} = \frac{1}{2} \sqrt{2(324+400)-1156} =$$

$$= \frac{1}{2} \sqrt{2 \cdot 724 - 1156} = \frac{1}{2} \sqrt{1448 - 1156} = \frac{1}{2} \sqrt{292} =$$

$$= \frac{1}{2} \sqrt{2 \cdot 2 \cdot 73} = \frac{2}{2} \sqrt{73} = \sqrt{73}$$

$$\begin{array}{r} 292 \\ 2 \\ \hline 146 \\ 2 \\ \hline 73 \end{array}$$

12. ABCD - трапеция

$$a = 10$$

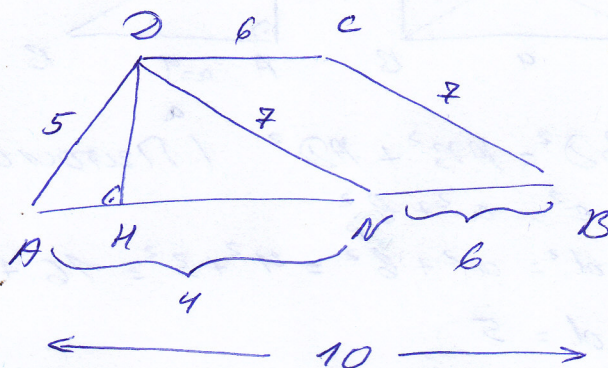
$$b = 6$$

$$c = 7$$

$$d = 5$$

$$S = ?$$

$$S = \frac{a+b}{2} h$$



Построение

$$DN \parallel BC$$

$$DN = BC = 7$$

$$AN = AB - BN$$

$$AN = 10 - 6 = 4$$

$$S_{AND} = \frac{1}{2} AN \cdot DH = \frac{1}{2} \cdot 4 \cdot h$$

$$S_{AND} = 2h \Rightarrow h = \frac{S_{AND}}{2}$$

$$S_{AND} = \sqrt{p(p-a)(p-b)(p-c)}$$

$$p = \frac{a+b+c}{2} = \frac{4+7+5}{2} = \frac{16}{2} = 8$$

$$S_{AND} = \sqrt{8(8-4)(8-7)(8-5)} = \sqrt{8 \cdot 4 \cdot 1 \cdot 3} = \sqrt{2 \cdot 4 \cdot 4 \cdot 3} = 4\sqrt{2 \cdot 3} = 4\sqrt{6}$$

$$\Rightarrow h = \frac{S_{AND}}{2} = \frac{4\sqrt{6}}{2} = 2\sqrt{6}$$

$$S = \frac{a+b}{2} h = \frac{10+6}{2} \cdot 2\sqrt{6} = 16\sqrt{6} \text{ см}^2$$

13. ABCD - трапеция

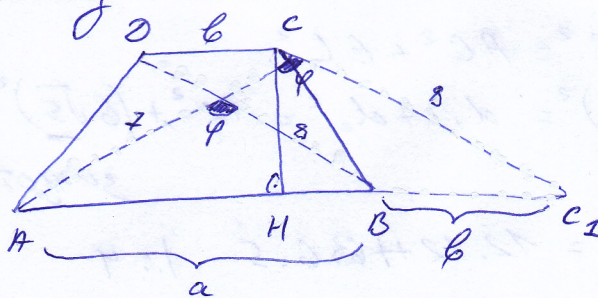
$$d_1 = 7$$

$$d_2 = 8$$

$$\varphi = 120^\circ$$

$$MN = ?$$

(среди ост.)



$$MN = \frac{a+b}{2}$$

$$\cos \varphi \Rightarrow c^2 = a^2 + b^2 - 2ab \cos \varphi$$

Постр. $CC_1 \parallel BD$

$$CC_1 = BD = 8$$

$$\Delta ACC_1 \quad AC_1^2 = AC^2 + CC_1^2 - 2AC \cdot CC_1 \cos \varphi = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos 120^\circ$$

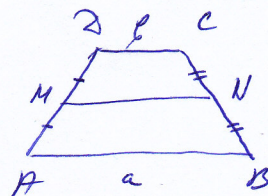
$$\cos 120^\circ = -\frac{1}{2}$$

$$AC_1^2 = 49 + 64 - 2 \cdot 7 \cdot 8 \cdot \left(-\frac{1}{2}\right) = 49 + 64 + 56 = 169 = 13^2$$

$$AC_1 = 13$$

$$AC_1 = AB + BC_1 = a + b = 13$$

$$MN = \frac{a+b}{2} = \frac{13}{2} = 6,5$$

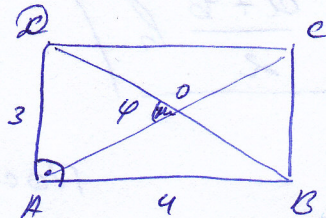


14з. ABCD - прямоугольник

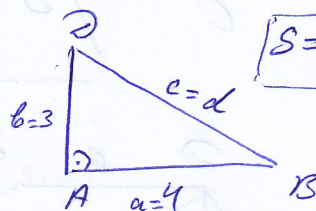
$$a = 4$$

$$b = 3$$

$$\cos \varphi = ?$$



$$S = a \cdot b = 3 \cdot 4 = 12$$



$$S = \frac{1}{2} d_1 \cdot d_2 \cdot \sin \varphi$$

$$\text{Озн. } d_1 = d_2 = d$$

$\triangle ABD$ - пр. $BD^2 = AB^2 + AD^2$ (Пифагорова (7))

$$c^2 = a^2 + b^2$$

$$\text{Т.е. } d^2 = a^2 + b^2 = 4^2 + 3^2 = 16 + 9 = 25 = 5^2$$

$$d = 5$$

$$S = \frac{1}{2} d \cdot d \cdot \sin \varphi = \frac{1}{2} d^2 \sin \varphi \Rightarrow \sin \varphi = \frac{2S}{d^2} = \frac{2 \cdot 12}{25} = \frac{24}{25}$$

$$\cos \varphi = \sqrt{1 - \sin^2 \varphi} = \sqrt{1 - \left(\frac{24}{25}\right)^2} = \sqrt{1 - \frac{576}{625}} = \sqrt{\frac{625 - 576}{625}} = \sqrt{\frac{49}{625}} = \frac{7}{25}$$

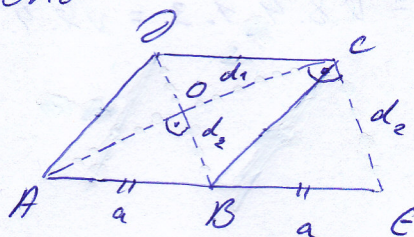
в формуле на д-к е $\oplus$ обш знаменател 625

15з. ABCD - ромб

$$d_1 = 12$$

$$d_2 = 6\sqrt{5}$$

$$a = ?$$



$$S = \frac{1}{2} d_1 \cdot d_2 \quad | \quad S = a \cdot h$$

$$\text{Озн. } AC = d_1, \quad BD = d_2$$

$$\text{Поскоп. } CE \parallel BD, \quad CE = BD = d_2$$

В ромба диагонали са перпендикулярни!

Т.е. $AC \perp BD$, но $CE \parallel BD \Rightarrow AC \perp CE \Rightarrow \triangle ACE$ - пр.

$$\triangle ACE - \text{пр.} \Rightarrow AE^2 = AC^2 + CE^2$$

$$(2a)^2 = d_1^2 + d_2^2 = 12^2 + (6\sqrt{5})^2 = 12 \cdot 12 + 6 \cdot 6 \cdot 5$$

$$\text{защото } \sqrt{5}^2 = 5!$$

$$4a^2 = 12 \cdot 12 + 36 \cdot 5 \quad | : 4$$

$$a^2 = 3 \cdot 12 + 9 \cdot 5 = 36 + 45 = 81 = 9^2 \Rightarrow a = 9$$

Правилни многоъгълници:

$$S_n = \frac{n R^2 \sin \frac{360^\circ}{n}}{2}$$

$$S_n = \frac{n a^2 \cotg \frac{180^\circ}{n}}{4}$$

$$S_n = n r^2 \tg \frac{180^\circ}{n}$$

където n = броят на страните.

16з. Правилник 6-ъг.
 $R = 5$

$$S = \frac{n R^2 \sin \frac{360^\circ}{n}}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$S_6 = ?$$

$$S_6 = \frac{6}{2} \cdot 5^2 \cdot \sin \frac{360^\circ}{6} = 3 \cdot 5^2 \cdot \sin 60^\circ = 3 \cdot 25 \cdot \frac{\sqrt{3}}{2} = \frac{75\sqrt{3}}{2} = 37.5\sqrt{3}$$

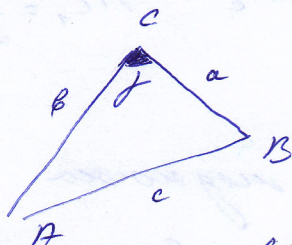
това са градуси!!! Използвайте право да съкратите
6-те (едната пред sin, другата в градусите!)

17з. $\triangle ABC$
 $a = 14$

$$b = 15$$

$$S = 84$$

$$c = ? \quad r = ?$$



$$S = \frac{1}{2} a \cdot b \cdot \sin \gamma \Rightarrow \sin \gamma = \frac{2S}{ab}$$

$$\cos \gamma = \pm \sqrt{1 - \sin^2 \gamma}$$

$$\cos P: c^2 = a^2 + b^2 - 2ab \cos \gamma$$

$$S = p \cdot r \Rightarrow r = \frac{S}{p}, \quad p = \frac{a+b+c}{2}$$

$$\sin \gamma = \frac{2S}{ab} = \frac{2 \cdot 84}{14 \cdot 15} = \frac{84}{7 \cdot 15} = \frac{2 \cdot 42}{7 \cdot 15} = \frac{2 \cdot 6}{15} = \frac{2 \cdot 2}{5} = \frac{4}{5}$$

$$\cos \gamma = + \sqrt{1 - \sin^2 \gamma} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{25-16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

$$c^2 = a^2 + b^2 - 2ab \cos \gamma = 14^2 + 15^2 - 2 \cdot 14 \cdot 15 \cdot \frac{3}{5} = 196 + 225 - 2 \cdot 14 \cdot 3 \cdot 3 = 196 + 225 - 252 = 169 = 13^2$$

$$c = 13$$

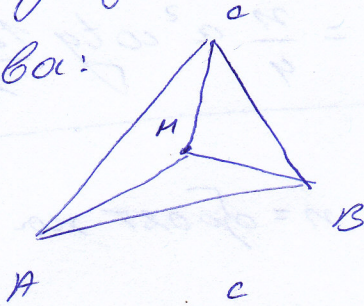
$$p = \frac{a+b+c}{2} = \frac{14+15+13}{2} = \frac{42}{2} = 21$$

$$S = p \cdot r$$

$$r = \frac{S}{p} = \frac{84}{21} = 4$$

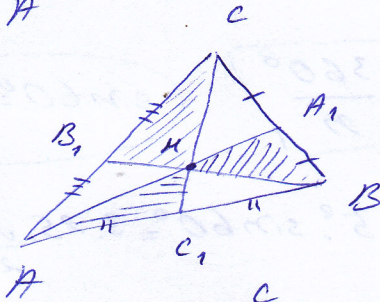
$\triangle ABC$
 m - медиантор (пресечна точка на медианите)

Свойства:



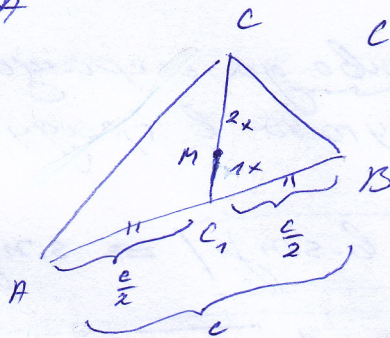
$$S_{ABM} = S_{BMC} = S_{AMC} = \frac{1}{3} S_{ABC} = \left(\frac{1}{3} S \right)$$

(3 големи равноликовреви $\triangle$ -ко)



$$S_{AMC_1} = S_{BMC_1} = S_{BMA_1} = S_{CMA_1} = S_{CMB_1} = S_{AMB_1} = \left(\frac{1}{6} S \right)$$

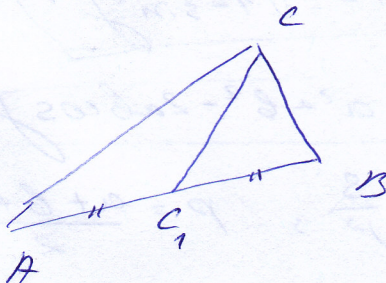
(6 малки равноликовреви $\triangle$ -ко)



$CC_1 = m_c = 3x$ - медиано

$$CM : MC_1 = 2 : 1$$

$$CM = \frac{2}{3} m_c \quad MC_1 = \frac{1}{3} m_c$$



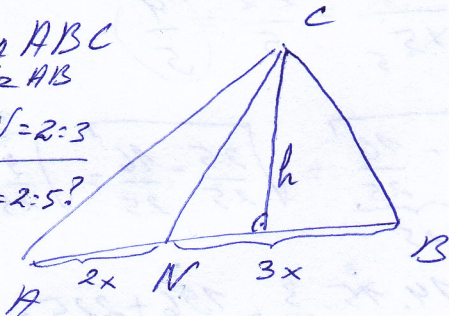
CC_1 - медиано

$$S_{ACC_1} = S_{BCC_1} = \left(\frac{1}{2} S \right)$$

163. $\triangle ABC$
 $N \in AB$

$$AN : BN = 2 : 3$$

$$S_{ANC} : S_{ABC} = 2 : 5?$$



$$\text{Озк. } AN = 2x, BN = 3x$$

$$S_{ABC} = \frac{1}{2} AB \cdot h = \frac{1}{2} \cdot 5x \cdot h = 5 \cdot \left(\frac{1}{2} x \cdot h \right) = 5y$$

$$\text{Положиме: } y = \frac{1}{2} x \cdot h$$

$$S_{ANC} = \frac{1}{2} AN \cdot h = \frac{1}{2} \cdot 2x \cdot h = 2 \cdot \left(\frac{1}{2} x \cdot h \right) = 2y$$

$$\frac{S_{ANC}}{S_{ABC}} = \frac{2y}{5y} = \frac{2}{5} \Rightarrow S_{ANC} : S_{ABC} = 2 : 5$$

Правилни многоъгълници:

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където n = броят на страните.

16з. Правилник 6-ъг. $R = 5$

$$S = \frac{n R^2 \sin \frac{360^\circ}{n}}{2}$$

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това са градуси!!! Износе право да съкратим
6-те (едно от пред sin, другото в градусите!)

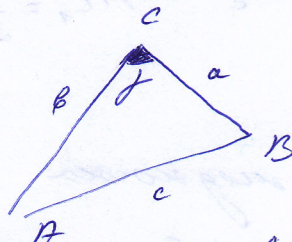
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$$\cos \gamma = \pm \sqrt{1 - \sin^2 \gamma}$$

$$\cos P: |c^2 = a^2 + b^2 - 2ab \cos \gamma|$$

$$S = p \cdot r \Rightarrow r = \frac{S}{p}, \quad p = \frac{a+b+c}{2}$$

$$\sin \gamma = \frac{2S}{ab} = \frac{2 \cdot 84}{14 \cdot 15} = \frac{84}{7 \cdot 15} = \frac{2 \cdot 42}{7 \cdot 15} = \frac{2 \cdot 6}{15} = \frac{2 \cdot 2}{5} = \frac{4}{5}$$

$$\cos \gamma = + \sqrt{1 - \sin^2 \gamma} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{25-16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

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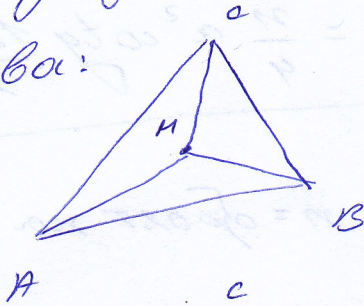
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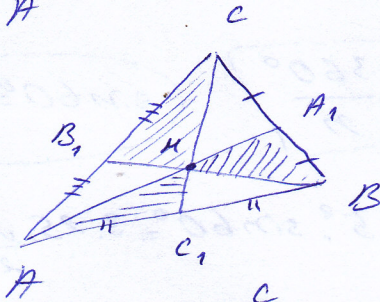
$\triangle ABC$
 M - медицентр (пересекна точка на медианите)

Свойства:



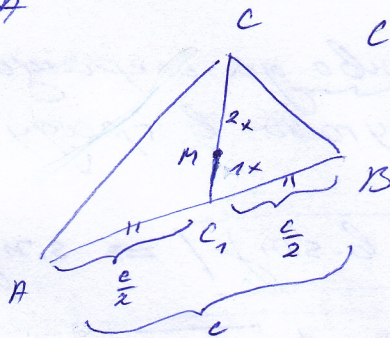
$$S_{ABM} = S_{BMC} = S_{AMC} = \frac{1}{3} S_{ABC} = \left(\frac{1}{3} S \right)$$

(3 голени равнолицеве $\triangle$ -ко)



$$S_{AMC_1} = S_{BMC_1} = S_{BMA_1} = S_{CMA_1} = S_{CMB_1} = S_{AMB_1} = \left(\frac{1}{6} S \right)$$

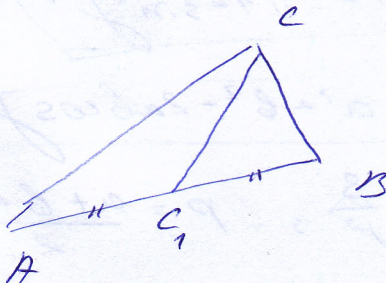
(6 малки равнолицеве $\triangle$ -ко)



$CC_1 = m_c = 3x$ - медиана

$$CM : MC_1 = 2 : 1$$

$$CM = \frac{2}{3} m_c \quad MC_1 = \frac{1}{3} m_c$$



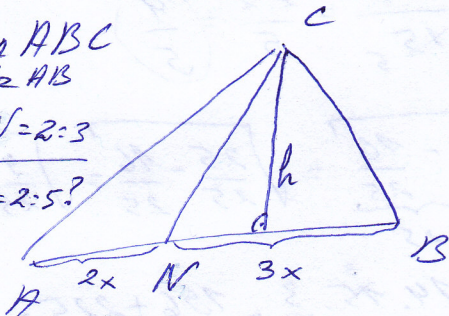
CC_1 - медиана

$$S_{ACC_1} = S_{BCC_1} = \left(\frac{1}{2} S \right)$$

163. $\triangle ABC$
 $N \in AB$

$$AN : BN = 2 : 3$$

$$S_{ANC} : S_{ABC} = 2 : 5?$$



$$\text{Озк. } AN = 2x, BN = 3x$$

$$S_{ABC} = \frac{1}{2} AB \cdot h = \frac{1}{2} \cdot 5x \cdot h = 5 \cdot \left(\frac{1}{2} x \cdot h \right) = 5y$$

$$\text{Положиме: } y = \frac{1}{2} x \cdot h$$

$$S_{ANC} = \frac{1}{2} AN \cdot h = \frac{1}{2} \cdot 2x \cdot h = 2 \cdot \left(\frac{1}{2} x \cdot h \right) = 2y$$

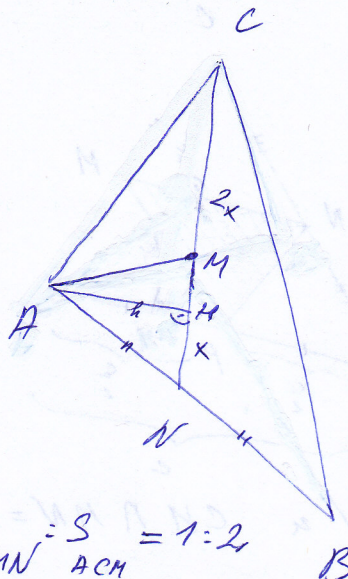
$$\frac{S_{ANC}}{S_{ABC}} = \frac{2y}{5y} = \frac{2}{5} \Rightarrow S_{ANC} : S_{ABC} = 2 : 5$$

17з. $\triangle ABC$

M - медицентър

N - среда на AB

$$S_{AMN} : S_{ACM} = 1 : 2 ?$$



Озн. $AM = h$, $AM \perp CN$

$$CM : MN = 2 : 1 = 2x : x$$

$$S_{AMN} = \frac{1}{2} MN \cdot AM = \frac{1}{2} x \cdot h = y$$

Полагаме $y = \frac{1}{2} x \cdot h$

$$S_{ACM} = \frac{1}{2} CM \cdot AM = \frac{1}{2} \cdot 2x \cdot h = 2 \cdot \left(\frac{1}{2} x \cdot h \right) = 2y$$

$$\Rightarrow \frac{S_{AMN}}{S_{ACM}} = \frac{y}{2y} = \frac{1}{2} \Rightarrow S_{AMN} : S_{ACM} = 1 : 2$$

18з. $\triangle ABC$

Да се помери $S_{\triangle}$ -х, построен от m_a, m_b, m_c .

M - медицентър, той дели медианите в отношение 2:1

$$\Rightarrow AM = \frac{2}{3} m_a, \quad MB = \frac{2}{3} m_b, \quad MC = \frac{2}{3} m_c$$

$$CC_1 = m_c, \quad CM = \frac{2}{3} m_c, \quad MC_1 = \frac{1}{3} m_c$$

За да построим $\triangle A_1 N_1 M$ със страни медианите, удовлетворява:

$$MC_1 = \frac{1}{3} m_c$$

$$C_1 N = \frac{1}{3} m_c$$

$$NN_1 = \frac{1}{3} m_c$$

$$MN = \frac{1}{3} m_b + \frac{1}{3} m_c = \frac{2}{3} m_c$$

$$MN_1 = \frac{1}{3} m_c + \frac{1}{3} m_c + \frac{1}{3} m_c = \frac{3}{3} m_c = m_c$$

$$AM = \frac{2}{3} m_a, \quad AA_1 = \frac{1}{3} m_a \Rightarrow MA_1 = \frac{2}{3} m_a + \frac{1}{3} m_a = \frac{3}{3} m_a = m_a$$

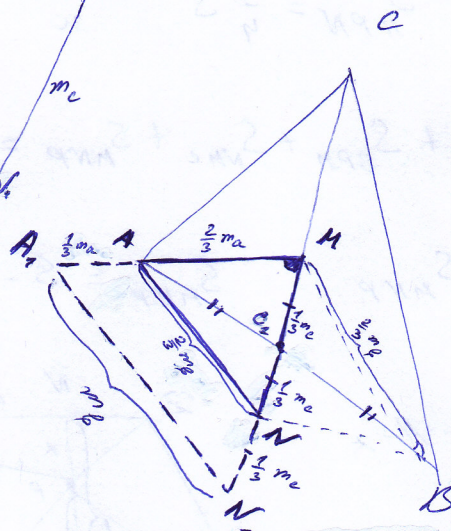
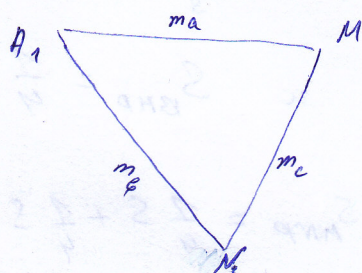
Полагаме $\triangle ANM$: $AM = \frac{2}{3} m_a$, $AN = BM = \frac{2}{3} m_b$ (успоредни) $MN = \frac{2}{3} m_c$

$$S = \frac{1}{2} a \cdot b \cdot \sin \angle$$

$$S_{ANM} = \frac{1}{2} AM \cdot MN \cdot \sin \angle M = \frac{1}{2} \cdot \frac{2}{3} m_a \cdot \frac{2}{3} m_c \cdot \sin \angle M = \frac{4}{9} \left(\frac{1}{2} m_a \cdot m_c \cdot \sin \angle M \right) = \frac{4}{9} y$$

$$S_{A_1 M N_1} = \frac{1}{2} A_1 M \cdot M N_1 \cdot \sin \angle M = \left(\frac{1}{2} m_a \cdot m_c \cdot \sin \angle M \right) = y$$

$$\Rightarrow S_{A_1 M N_1} = \frac{y}{\frac{4}{9} y} = \frac{9}{4} S_{ANM} = \frac{9}{4} \cdot \frac{4}{9} S = S$$



Озн. $S_{ABC} = S$

$$S_{A_1 M} = \frac{1}{3} S$$

$$S_{C_1 B M} = \frac{1}{6} S$$

но $\triangle C_1 B M \cong \triangle A C_1 N$

+ C_1 - връзки

$$AC_1 = BC_1 = \frac{c}{2}$$

$$MC_1 = NC_1 = \frac{1}{3} m_c$$

$$\Rightarrow S_{ANM} = S_{BNC_1} = \frac{1}{6} S$$

$$S_{ANM} = \frac{1}{6} S + \frac{1}{6} S = \frac{2}{6} S = \frac{1}{3} S$$

19. $\triangle ABC$

$$S_{ABC} = S$$

M, N, P - середины BC, CA, AB

$$S_{MNP} = ?$$

CH - высота

$$CH \perp AB \Rightarrow CH \perp MN \text{ и } CH \cap MN = F, \quad CF = \frac{h}{2}$$

$$S_{MNC} = \frac{1}{2} MN \cdot CF = \frac{1}{2} \cdot \left(\frac{c}{2}\right) \cdot \left(\frac{h}{2}\right) = \frac{1}{4} \cdot \left(\frac{1}{2} c \cdot h\right) = \frac{1}{4} S$$

аналогично $S_{APN} = \frac{1}{4} S$ и $S_{BMP} = \frac{1}{4} S$

$$S_{ABC} = S_{APN} + S_{BMP} + S_{MNC} + S_{MNP} = \frac{1}{4} S + \frac{1}{4} S + \frac{1}{4} S + S_{MNP} = \frac{3}{4} S + S_{MNP}$$

$$S = \frac{3}{4} S + S_{MNP}$$

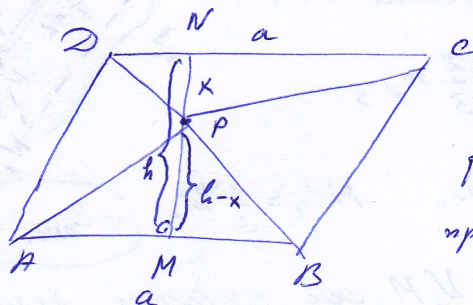
$$S_{MNP} = S - \frac{3}{4} S = \frac{4}{4} S - \frac{3}{4} S = \frac{1}{4} S$$

20. $ABCD$ - Π -к

$$S_{ABCD} = S$$

P - внутренняя т.к.

$$S_{APB} + S_{CPD} = \frac{1}{2} S ?$$



$$S_{ABCD} = h \cdot a \quad | S = h \cdot a |$$

Построение через P
прямая $MN \perp AB$: $\begin{cases} PN = x \\ MN = h \end{cases} \Rightarrow PM = h - x$

$$S_{APB} = \frac{1}{2} AB \cdot PM = \frac{1}{2} a \cdot (h - x) \quad S_{CPD} = \frac{1}{2} CD \cdot PN = \frac{1}{2} a \cdot x$$

$$S_{APB} + S_{CPD} = \frac{1}{2} a \cdot (h - x) + \frac{1}{2} a \cdot x = \frac{1}{2} a (h - x + x) = \frac{1}{2} a \cdot h = \frac{1}{2} S$$

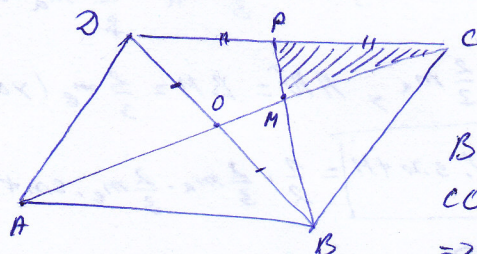
21. $ABCD$ - Π -к

$$S_{ABCD} = S$$

P - середина на CD

$BP \cap AC = M$

$$S_{MBP} = S = 1:12 ?$$



$$S_{BCD} = \frac{1}{2} S$$

BP - медиана в $\triangle BCD$

CO - медиана в $\triangle BCD$ (O - середина диаг.)
 $\Rightarrow M$ - медиантер т.к. $BP \cap CO = M$

$$\Rightarrow S_{MBP} = \frac{1}{6} S_{BCD} = \frac{1}{6} \cdot \frac{1}{2} S = \frac{1}{12} S$$